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Saveetha Institute of Medical And Technical Sciences

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Technical Presentation • CSA - 1409 Compiler Design

APPLICATIONS OF COMPILERS

How compiler technology powers software and modern computing



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1. Introduction to Compilers

What does a compiler do?

- A compiler translates a program written in a high-level language into a target form, often machine code.
- It helps the computer execute instructions written in languages such as C, C++ and Rust.
- Compiler stages commonly include lexical analysis, parsing, semantic analysis, optimization and code generation.
- Compilers also report errors and can improve program performance.
- Compiler technology is used in many areas beyond translating source code.

2. How a Compiler Works

A simplified compiler pipeline



Optimization may improve the generated code before the final output.

Input: source program → Output: target code (or another target representation)

3. Application in Software Development

- Compilers convert developers' code into executable programs.
- Used to build desktop software, command-line tools and system utilities.
- Examples: Desktop applications, calculators, and other software
- Compiler warnings and errors help developers find problems early.
- Optimization can reduce execution time or memory use when applied appropriately.

4. Operating Systems and System Software

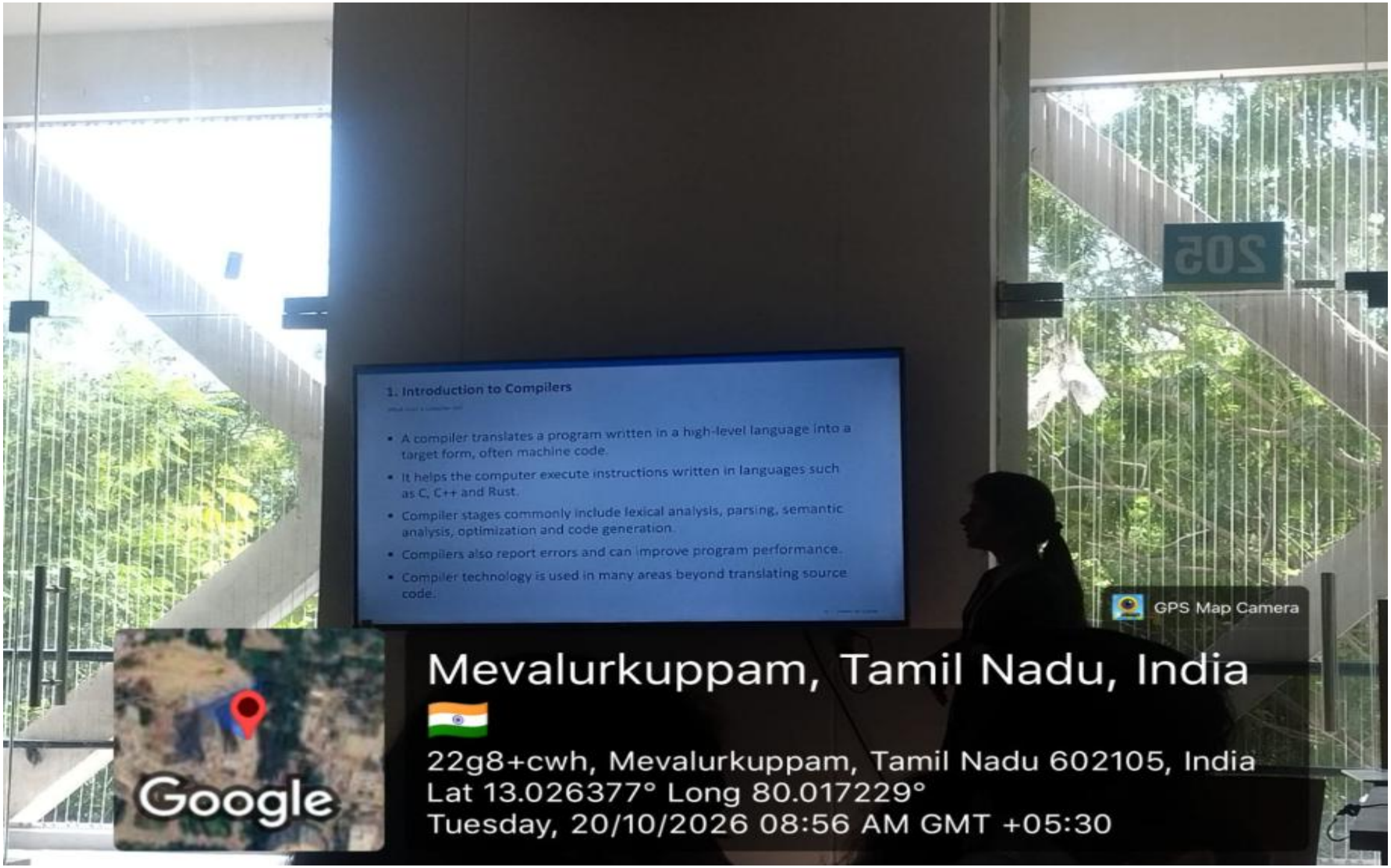
- Operating-system components and device drivers are often written in compiled languages.
- Compilers help produce low-level code that interacts with hardware.
- Used in kernels, embedded software, networking tools and performance-critical libraries.
- Cross-compilers can build software for a different processor or device than the computer running the compiler.
- Example: compiling code on a laptop for an ARM-based embedded board.

5. Mobile, Embedded and IoT Systems

- Compilers help create software for phones, smart devices, vehicles and sensors.
- Embedded devices often have limited memory, power and processing capacity.
- Compiler optimizations can help reduce code size and improve performance.
- Cross-compilation is common when the target device cannot compile its own software.
- Examples: firmware for microcontrollers and software for smart appliances.

6. Web, Cloud and Managed Runtimes

- Some languages compile to an intermediate representation or bytecode.
- Java source code is compiled into bytecode, which runs on the Java Virtual Machine (JVM).
- Web tools can compile languages such as TypeScript to JavaScript.
- Just-in-time (JIT) compilers translate frequently used code while a program is running.
- These approaches support portability, runtime optimization and large-scale applications.


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• Compilers also report errors and can improve program performance.
• Compiler technology is used in many areas beyond translating source code.
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